

1 From forecasting to real-time surveillance: An
2 anytime-valid monitoring framework for technology
3 diffusion, with 35 years of evidence from the United
4 States

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6 **Abstract**

Technological forecasting has traditionally produced a curve fitted after data collection ends, leaving planners without a disciplined way to ask, in real time, whether a technology's diffusion has actually changed regime. We reframe forecasting as continuous surveillance: an anytime-valid monitoring framework for technology diffusion that combines a causal measurement layer, treating survey and instrument redesigns as interventions on the observation process, with one-sided mixture e-processes whose error control holds uniformly over time. Applied to the U.S. Census Bureau's biweekly AI-adoption survey, the monitor declares a diffusion takeoff on 20 April 2025, quantifies the mid-2025 deceleration as sub-threshold evidence later vindicated by re-acceleration, and correctly absorbs a 26-standard-error November 2025 question-wording redesign as measurement, unlike conventional repeated tests. A retrospective extension recovers three decades of internet and e-commerce diffusion shifts.

7 *Keywords:* technology diffusion, artificial intelligence adoption,
8 anytime-valid inference, e-values, sequential monitoring, survey

10 1. Introduction

11 Technological forecasting has long been understood as a planning tool: a
 12 discipline for anticipating the emergence, growth, and impact of new tech-
 13 nologies so that managers and policymakers can act ahead of, rather than be-
 14 hind, the curve (Martino, 2003). Diffusion forecasting in particular—fitting
 15 growth curves to historical adoption data—has a quarter-century of method-
 16 ological development and review (Meade and Islam, 2006). Yet nearly all of
 17 that tradition treats forecasting as something done *after* a data collection
 18 window closes, on a fixed sample, once. It offers no answer to a question
 19 planners now face continuously: how can a manager, analyst, or policymaker
 20 know—with stated statistical confidence, and at the moment it matters—
 21 that the diffusion of a new technology has changed regime: that adoption
 22 has taken off, that growth has stalled, or that an apparent shock is an artifact
 23 of how adoption is measured? The question has become concrete with arti-
 24 ficial intelligence. Since September 2023 the U.S. Census Bureau’s Business
 25 Trends and Outlook Survey (BTOS) has asked a rotating, nationally rep-
 26 resentative sample of roughly 200,000 employer businesses every two weeks
 27 whether they used AI, producing the highest-frequency official adoption se-
 28 ries ever published for a general-purpose technology (Bonney et al., 2024).
 29 Firms maintain comparable internal telemetry on their own AI deployments
 30 that updates even faster. Adoption evidence now arrives as a stream, and it
 31 is consumed as one: dashboards are refreshed at every release, and decisions
 32 to scale, hold, or stop technology investments are implicitly revisited at each

33 refresh.

34 The inferential toolkit of technology and innovation management (TIM)
35 research was not built for this regime. The diffusion literature—from the
36 epidemic and probit traditions (Bass, 1969; Geroski, 2000; Rogers, 2003) to
37 contemporary firm-level studies of AI uptake (McElheran et al., 2024; Zolas
38 et al., 2020)—estimates adoption curves retrospectively and tests hypotheses
39 once, on a closed sample. When the same tests are applied repeatedly to an
40 accumulating series, as any monitoring practice implicitly does, their guar-
41 antees dissolve: the probability of at least one spurious “significant” finding
42 grows without bound in the number of looks, a fact established in sequential
43 analysis decades ago (Wald, 1945; Armitage et al., 1969) and rediscovered
44 in industrial online experimentation as the “peeking problem” (Johari et al.,
45 2022). In this paper’s empirical setting the problem is not hypothetical: we
46 show that a standard rolling-window test applied to the BTOS AI series at
47 every release generates alarms on 37 of 48 possible looks, and on 31 of 48
48 even after Bonferroni correction, rendering “significance” useless as a signal
49 of regime change.

50 A second obstacle is that high-frequency adoption measurement is itself
51 unstable. On 17 November 2025 the Census Bureau replaced the BTOS
52 question on AI use “in producing goods or services” with AI use in “any of
53 its business functions.” The published national adoption rate moved from
54 10.0 percent in the last legacy-wording wave to 17.3 percent in the first
55 revised-wording wave—a jump of 7.3 percentage points, or 25.7 design-based
56 standard errors, that dwarfs every genuine movement in the series and re-
57 flects wording, not diffusion (Bick, 2026). Any credible monitoring scheme

58 must separate shifts in the *latent diffusion state* from interventions on the
59 *measurement apparatus*; the distinction is inherently causal (Pearl, 2009),
60 and purely statistical change detectors cannot draw it.

61 We address both obstacles jointly and ask three research questions:

- 62 • **RQ1.** Can regime shifts in technology diffusion be detected in near-
63 real time with error guarantees that remain valid under continuous,
64 open-ended monitoring?
- 65 • **RQ2.** How can genuine shifts in the latent diffusion state be dis-
66 tinguished, in real time, from changes in the survey instrument that
67 measures it?
- 68 • **RQ3.** What does such a monitor reveal about the dynamics of digi-
69 tal technology diffusion in the United States—with enterprise AI over
70 2023–2025 as the high-frequency focal case, set against the long view
71 since 1990—and how does its behavior compare with the repeated-
72 testing practices embedded in current monitoring conventions?

73 Our framework has two components. The first is a causal measurement
74 layer: the observed adoption rate is generated by a latent diffusion state
75 together with an explicit measurement-regime variable, and instrument re-
76 designs are modeled as interventions—in the do-operator sense—on the mea-
77 surement regime alone. Pre-announced redesign waves contribute no evidence
78 about diffusion, and a cross-strata diagnostic rooted in invariance logic (Pe-
79 ters et al., 2016; Rothenhäusler et al., 2021) verifies the attribution: an
80 instrument event moves all strata simultaneously in a single wave, whereas
81 genuine diffusion shifts build gradually and heterogeneously. The second

82 component is an e-process monitor. E-values and safe anytime-valid infer-
 83 ence (Shafer and Vovk, 2019; Vovk and Wang, 2021; Howard et al., 2021;
 84 Ramdas et al., 2023; Grünwald et al., 2024) provide nonnegative processes
 85 that are martingales under the null hypothesis of “no departure from the
 86 baseline adoption path,” so that by Ville’s inequality the probability that
 87 the process *ever* crosses $1/\alpha$ under the null is at most α , regardless of how
 88 often the analyst looks or when monitoring stops. We instantiate the moni-
 89 tor with Robbins-type one-sided normal-mixture e-processes (Robbins, 1970)
 90 on adoption increments standardized by the survey’s published design-based
 91 standard errors, organized into surveillance epochs with re-anchoring after
 92 each declared shift, in the spirit of sequential change detection (Page, 1954;
 93 Shin et al., 2023).

94 Applying the framework to all 56 published BTOS waves through Decem-
 95 ber 2025, and retrospectively to three decades of U.S. digital diffusion, yields
 96 four substantive findings. First, the monitor declares a diffusion takeoff on
 97 20 April 2025 with terminal e-value 42.2, after an evidence trajectory that
 98 surged to 15.4 by 17 December 2023 and reached a local peak of 17.1 on
 99 28 January 2024, collapsed to 0.4 by 7 April 2024 during the first-half-2024
 100 growth stall (when measured adoption slipped from 4.9 to 4.7 percent), and
 101 rebuilt through the late-2024 and early-2025 acceleration. Second, against
 102 the anchored early-2025 growth rate of 0.299 percentage points per wave,
 103 the downward monitor quantifies the widely discussed mid-2025 decelera-
 104 tion at a peak evidence ratio of 24.3-to-1 on 7 September 2025—a strong
 105 “amber” that never crosses the declaration threshold; the restraint is vindi-
 106 cated within weeks as growth resumes to 10.0 percent, and it contrasts with

107 a moving-average heuristic that whipsaws twice through the same episode.
108 Third, the November 2025 redesign, the largest movement in the series by an
109 order of magnitude, is excluded by design and confirmed as a measurement
110 event by its signature: synchronized single-wave jumps of 6.1 to 13.8 stan-
111 dard errors across all seven employment-size classes, against within-regime
112 95th-percentile movements of 1.7 to 2.9. Fourth, a retrospective application
113 of the same surveillance logic—with a predictable plug-in variance in place of
114 design-based standard errors—recovers the punctuated evolutionary record
115 of U.S. digital diffusion since 1990: on the annual internet-penetration se-
116 ries it declares the takeoff in 1995 and the onset of saturation in 2008, and
117 on the quarterly e-commerce share of retail sales it declares the takeoff in
118 2004Q1, a mid-2010s acceleration in 2018Q1, the COVID-19 shock in the
119 very quarter it occurred (2020Q2), and the post-pandemic re-acceleration in
120 2023Q1, while flagging a single-year 2016 discontinuity in the internet se-
121 ries whose provenance cannot be settled without measurement metadata—a
122 long-view illustration of exactly why the causal measurement layer matters.
123 A companion timeline (Figure 7) aligns the intermediate technology waves
124 of the analytics era—data mining, cloud computing, big data, data science,
125 and deep learning—with the measured record, dated by canonical landmarks
126 because no official penetration series were ever produced for them.

127 The paper contributes to TIM methodology by importing anytime-valid
128 inference into diffusion research, replacing repeated testing with a procedure
129 whose error control survives the way monitoring is actually practiced; to
130 diffusion theory by formalizing survey redesigns as causal interventions on
131 the observation process, converting a measurement-comparability nuisance

132 (Crane et al., 2025) into an identified model component; and to practice by
133 distilling the machinery into a decision protocol—graded evidence thresholds
134 mapped to scale/hold/stop actions and re-anchoring rules—that applies as
135 readily to a firm’s internal adoption telemetry as to official statistics. In the
136 vocabulary of this special issue, the framework operationalizes an evolution-
137 ary view of industry dynamics: the punctuated uptake of a social technology
138 is tracked as it unfolds, as a sequence of dated, error-controlled regime dec-
139 larations rather than a curve fitted in hindsight.

140 Section 2 reviews the relevant literatures. Section 3 develops the method-
141 ology. Section 4 describes the data. Section 5 reports results, Section 6
142 discusses implications and limitations, and Section 7 concludes.

143 2. Theoretical background and related literature

144 2.1. *Measuring technology diffusion: from annual curves to biweekly streams*

145 Classical diffusion research models cumulative adoption as a smooth curve—
146 logistic or Bass-type under epidemic assumptions (Bass, 1969), or the aggre-
147 gate of heterogeneous threshold decisions in probit-style models (Geroski,
148 2000; Rogers, 2003)—while firm-level adoption studies in the technology-
149 acceptance tradition explain cross-sectional variation in who adopts (Davis,
150 1989). These literatures share a data regime: adoption observed annually
151 or less often, on closed samples, with inference conducted once after data
152 collection ends.

153 AI has broken that regime. The BTOS delivers a biweekly, nationally
154 representative firm-level adoption series (Bonney et al., 2024); the Annual
155 Business Survey provides deeper but slower benchmarks (Zolas et al., 2020;

156 [McElheran et al., 2024](#)); and household surveys track worker-side uptake of
157 generative AI at high frequency ([Bick et al., 2025](#)). Substantively, adoption
158 is fast and uneven: BTOS-based studies document use rates that more than
159 doubled within two years of the series’ launch, strong firm-size gradients,
160 and sectoral concentration in Information, Finance, and Professional Services
161 ([U.S. Census Bureau, 2026a](#); [Garcia Luna, 2026](#)). Comparisons across instru-
162 ments reveal wide dispersion in point estimates—from about 5 to 40 percent
163 depending on question framing, respondent, and sampling unit ([Crane et al.,](#)
164 [2025](#))—and the November 2025 BTOS redesign showed that wording alone
165 can nearly double a measured national adoption rate ([Bick, 2026](#); [Federal](#)
166 [Reserve Board, 2026](#)). The literature has treated these measurement dis-
167 crepancies as a comparability nuisance to be footnoted; we treat them as
168 first-class objects in a causal model of the observation process.

169 *2.2. The peeking problem in adoption monitoring*

170 When an analyst inspects an accumulating series at every new wave and
171 applies a fixed-horizon test—rolling-window mean comparisons, regression
172 breaks, control-chart rules—each inspection is an implicit hypothesis test,
173 and the family-wise error rate compounds. [Armitage et al. \(1969\)](#) showed
174 that repeated significance testing at nominal level 0.05 produces a true false-
175 positive probability that grows without bound in the number of looks; [Johari](#)
176 [et al. \(2022\)](#) documented the same phenomenon in industrial A/B testing and
177 introduced always-valid p -values as a remedy. Group-sequential corrections
178 from clinical trials require the number and timing of looks to be fixed in
179 advance, an assumption that fits neither dashboards nor a survey program
180 whose supplements and redesigns arrive unpredictably. Technology-adoption

181 monitoring sits squarely in the uncorrected regime: to our knowledge, no
 182 published study of AI diffusion adjusts its inferences for the sequential, open-
 183 ended nature of the underlying data collection.

184 2.3. *E-values and anytime-valid inference*

185 E-values provide a measure of evidence against a null hypothesis with a
 186 betting interpretation: an e-variable is a nonnegative random variable with
 187 expectation at most one under the null, and an e-process is a nonnegative
 188 process upper-bounded by a supermartingale with initial value one under
 189 the null (Shafer and Vovk, 2019; Vovk and Wang, 2021; Ramdas et al.,
 190 2023; Grünwald et al., 2024). Ville’s inequality then yields time-uniform
 191 error control: $\mathbb{P}_{H_0}(\sup_t E_t \geq 1/\alpha) \leq \alpha$. Evidence accumulates multiplica-
 192 tively, monitoring may stop or continue at any data-dependent time, and
 193 rejecting when $E_t \geq 1/\alpha$ controls the type-I error at α *uniformly over time*.
 194 Concrete constructions include Robbins’s method of mixtures for Gaussian
 195 observations (Robbins, 1970), time-uniform confidence sequences (Howard
 196 et al., 2021), and betting-based procedures for bounded random variables
 197 (Waudby-Smith and Ramdas, 2024); e-process analogues of CUSUM sup-
 198 port sequential changepoint detection with restarts (Page, 1954; Shin et al.,
 199 2023). These tools were developed for clinical monitoring and online exper-
 200 imentation; their application to macro-level technology diffusion appears to
 201 be new.

202 2.4. *Causal structure of the measurement process*

203 Structural causal models (Pearl, 2009) distinguish changes in a system’s
 204 mechanisms from changes in how the system is observed. Invariant causal

205 prediction (Peters et al., 2016) and anchor regression (Rothenhäusler et al.,
 206 2021) operationalize a related idea: relationships that are structural re-
 207 main invariant across environments and interventions, whereas artifacts shift
 208 with the environment. In our setting the “environments” are measurement
 209 regimes—question wordings, panel compositions, supplement windows—and
 210 the invariance logic runs in a diagnostic direction: a genuine diffusion shift
 211 should be visible *within* a fixed measurement regime and build heteroge-
 212 neously *across* strata according to their exposure to the underlying driver,
 213 while a measurement intervention produces a discontinuity synchronized with
 214 the redesign date and common to all strata in a single wave. Section 3 em-
 215 beds this logic in the monitor itself, so that evidence accrued by the e-process
 216 concerns the diffusion state, not the instrument.

217 3. Methodology

218 Figure 1 summarizes the framework. A biweekly adoption stream feeds a
 219 causal measurement layer that excludes instrument-intervention waves and
 220 standardizes within-regime increments; one-sided mixture e-processes accu-
 221 mulate anytime-valid evidence against a baseline-path null; threshold cross-
 222 ings feed a managerial decision protocol; and each declared shift triggers
 223 re-anchoring of the monitor for a new surveillance epoch.

224 3.1. Observation model and null hypotheses

225 Let $t = 1, 2, \dots$ index biweekly survey waves within a measurement
 226 regime. The published series delivers a weighted adoption-rate estimate $\hat{\pi}_t$
 227 (percent of businesses reporting AI use in the past two weeks) with design-
 228 based standard error $\hat{\sigma}_t$, together with parallel series for planned adoption

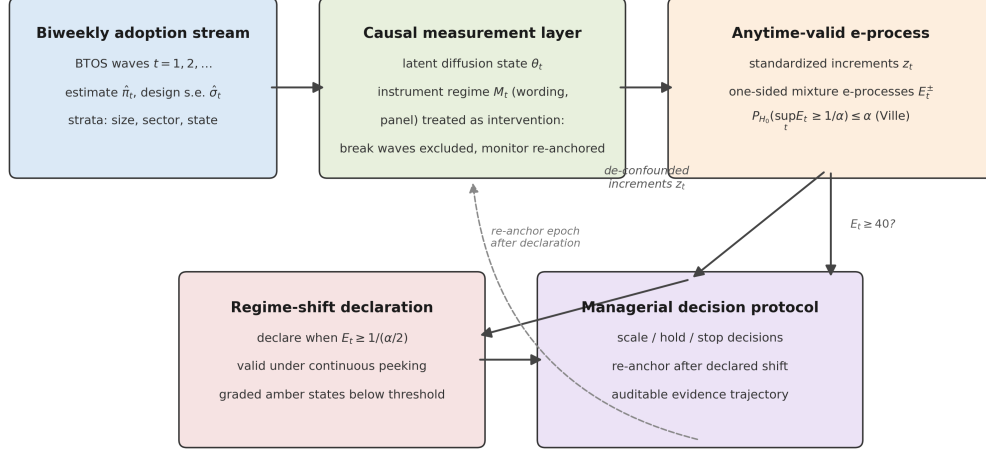


Figure 1: The causal e-process monitoring framework. The measurement layer separates the latent diffusion state θ_t from the instrument regime M_t ; one-sided mixture e-processes accumulate time-uniformly valid evidence on standardized within-regime increments; declared shifts and instrument changes trigger re-anchoring.

229 and for strata $k = 1, \dots, K$ (employment-size classes, sectors, states). We
 230 posit a latent diffusion state θ_t —the population adoption propensity under a
 231 fixed, ideal instrument—and a measurement regime M_t recording the active
 232 question wording and panel design. The observation equation is

$$\hat{\pi}_t = g(\theta_t, M_t) + \varepsilon_t, \quad \varepsilon_t \mid \mathcal{F}_{t-1} \sim \mathcal{N}(0, \hat{\sigma}_t^2), \quad (1)$$

233 where g maps the latent state through the instrument (a narrower wording
 234 truncates θ_t to its production-function component) and the Gaussian ap-
 235 proximation for the sampling error ε_t is justified by the survey’s scale: each
 236 biweekly estimate aggregates roughly 15,000–20,000 weighted responses (Fed-
 237 eral Reserve Board, 2026), and consecutive waves are drawn from different

rotation panels, so sampling errors are independent across waves. Instrument redesigns are interventions $\text{do}(M_t = m')$ on the observation process: they change g , not θ_t . The monitoring question is whether θ_t has departed from a baseline path—not whether $\hat{\pi}_t$ has jumped.

Within a regime, define the increment $\Delta_t = \hat{\pi}_t - \hat{\pi}_{t-1}$. A surveillance epoch e is endowed with a *baseline slope* β_e (percentage points per wave), fixed at the epoch’s start from information available at that time. The epoch’s null hypothesis is that the latent path continues at the baseline rate:

$$H_0^{(e)} : \quad \mathbb{E}[\Delta_t \mid \mathcal{F}_{t-1}] = \beta_e \quad \text{with} \quad \Delta_t - \beta_e \sim \mathcal{N}(0, \sigma_t^2), \quad \sigma_t^2 = \hat{\sigma}_t^2 + \hat{\sigma}_{t-1}^2 + \text{se}(\hat{\beta}_e)^2, \quad (2)$$

where $\text{se}(\hat{\beta}_e)$ accounts for estimation error in an anchored slope (zero when β_e is pre-registered rather than estimated). Under $H_0^{(e)}$ the standardized increments $z_t = (\Delta_t - \beta_e)/\sigma_t$ are independent standard normal. Rejecting $H_0^{(e)}$ upward is evidence of takeoff or acceleration relative to the baseline; rejecting downward is evidence of deceleration or stagnation. The null is deliberately sharp: it attributes all within-regime dispersion to sampling error, so that *any* persistent movement of the latent path away from the baseline counts as signal. Section 5.8 examines robustness to relaxing this attribution.

3.2. One-sided mixture e -processes

For each epoch we run two one-sided monitors built from Robbins’s method of mixtures (Robbins, 1970). Let $S_n = \sum_{s=1}^n z_s$ denote the cumulative standardized evidence after n within-epoch increments and let $\tau > 0$ be a prior scale. Mixing the Gaussian likelihood ratio $\exp(\delta S_n - n\delta^2/2)$ over

260 a half-normal prior on the drift $\delta > 0$ gives, in closed form (derivation in
 261 [Appendix A](#)),

$$E_n^+ = \frac{2}{\tau\sqrt{a_n}} \Phi\left(\frac{S_n}{\sqrt{a_n}}\right) \exp\left(\frac{S_n^2}{2a_n}\right), \quad a_n = n + \tau^{-2}, \quad (3)$$

262 with $E_0^+ = 1$, and symmetrically E_n^- obtained by replacing S_n with $-S_n$.
 263 Under $H_0^{(e)}$, (E_n^+) and (E_n^-) are nonnegative martingales with initial value
 264 one; more generally, under the composite one-sided null $\mathbb{E}[z_t \mid \mathcal{F}_{t-1}] \leq 0$
 265 (with unit-variance Gaussian noise), (E_n^+) is a supermartingale ([Appendix](#)
 266 [A](#)). Ville’s inequality therefore gives the time-uniform guarantee

$$\mathbb{P}_{H_0}\left(\exists n : E_n^+ \geq \frac{1}{\alpha/2}\right) \leq \frac{\alpha}{2}, \quad (4)$$

267 and likewise for E_n^- , so that declaring a regime shift the first time *either*
 268 monitor reaches $1/(\alpha/2) = 40$ (at $\alpha = 0.05$) controls the per-epoch false-
 269 declaration probability at α no matter how often the series is inspected or
 270 when monitoring stops. Throughout we set $\tau = 1$, a unit prior scale on
 271 the standardized drift; Section 5.8 shows the results are insensitive to this
 272 choice. Because E_n is a likelihood-ratio-type quantity, intermediate values
 273 carry graded meaning— $E_n = 10$ is 10-to-1 evidence against the baseline—
 274 which motivates the “amber” states of the decision protocol below, an in-
 275 terpretive resource that repeated p -values cannot supply under continuous
 276 monitoring.

277 3.3. Surveillance epochs and re-anchoring

278 Monitoring proceeds through epochs, in the restart spirit of CUSUM
 279 ([Page, 1954](#)) and its e-process descendants ([Shin et al., 2023](#)):

- 280 1. **Epoch 1 (takeoff surveillance).** The baseline slope is pre-registered
 281 at $\beta_1 = 0$: the null is that the latent adoption path is flat. Both
 282 one-sided monitors run from the first increment.
- 283 2. **Declaration.** The first time $E_n^+ \geq 40$ or $E_n^- \geq 40$, a regime shift is
 284 declared with its direction, date, and terminal e-value recorded.
- 285 3. **Re-anchoring.** The next epoch’s baseline slope $\hat{\beta}_{e+1}$ and its standard
 286 error are estimated by weighted least squares (weights $1/\hat{\sigma}_t^2$) on the
 287 $w = 8$ waves ending at the declaration wave—a window that uses only
 288 past data, so predictability and hence the supermartingale property
 289 are preserved. The new epoch begins at the following wave with fresh
 290 monitors at wealth one.
- 291 4. **Instrument interventions.** At a pre-announced redesign wave ($M_t \neq$
 292 M_{t-1}), the transition increment is excluded from every monitor—the
 293 do-operator made operational—and the monitor re-anchors on post-
 294 redesign waves once w of them have accumulated. For discontinuities
 295 that are *not* pre-announced, a cross-strata diagnostic (Section 3.4) al-
 296 locates the wave to the measurement layer or the diffusion layer before
 297 any evidence is booked.

298 3.4. *Cross-strata attribution diagnostic*

299 Let $z_{k,t}$ be the standardized increment in stratum k at wave t . The di-
 300 agnostic compares the profile of a candidate break wave against the stratum’s
 301 own within-regime history: an instrument event exhibits (i) synchronization—
 302 every stratum jumps in the same wave; (ii) extremity—each stratum’s $|z_{k,t}|$
 303 far exceeds the upper tail (we report the 95th percentile) of its within-regime

304 $|z|$ distribution; and (iii) sign coherence consistent with the mechanical ef-
 305 fect of the instrument change (a broader wording can only raise measured
 306 adoption). A genuine diffusion shift, by contrast, builds over multiple waves
 307 with magnitudes inside or near the within-regime tail and loads heteroge-
 308 neously on strata according to exposure. The diagnostic is a screening rule
 309 with a causal-invariance rationale (Peters et al., 2016; Rothenhäusler et al.,
 310 2021) rather than a formally identified estimator; in the application it is
 311 unambiguous (Section 5.3).

312 3.5. Retrospective surveillance with plug-in variance

313 The BTOS application enjoys a luxury most diffusion series lack: pub-
 314 lished design-based standard errors that make the sampling distribution
 315 of increments known. To extend surveillance to historical series without
 316 such metadata (Section 4.2), we replace σ_t in (2) with a *predictable plug-in*
 317 scale: $\hat{\sigma}_t = \max\{\phi, 1.4826 \cdot \text{MAD}(r_{t-W}, \dots, r_{t-1})\}$, the median-absolute-
 318 deviation estimate over the trailing $W = 8$ within-epoch increment residuals
 319 $r_s = \Delta_s - \beta_e$, seeded at each re-anchoring from the anchoring window’s own
 320 residuals and floored at a series-specific constant ϕ chosen conservatively
 321 (0.8 percentage points per year for the annual internet series; 0.08 points per
 322 quarter for the quarterly e-commerce series). Because the scale at wave t uses
 323 only information through $t - 1$, predictability—and hence the supermartin-
 324 gale structure—is preserved mechanically; but the Gaussian null in (2) now
 325 carries an estimated rather than known variance, so the time-uniform guaran-
 326 tee (4) holds only approximately. We therefore treat the retrospective results
 327 as an illustrative application of the surveillance logic, report e-values above
 328 10^4 simply as decisive, and reserve exact error statements for the design-based

Table 1: Decision protocol linking e-process states to managerial actions ($\alpha = 0.05$; declaration threshold $1/(\alpha/2) = 40$ per one-sided monitor).

Monitor state	Evidence level	Interpretation	Illustrative action
$E_n < 2$	Negligible	No credible departure from baseline path	Continue routine monitoring
$2 \leq E_n < 10$	Amber	Suggestive evidence; odds 2–10:1 against baseline	Prepare contingency; increase review cadence
$10 \leq E_n < 40$	Strong amber	Odds 10–40:1; declaration plausible soon	Pre-position decision; brief stakeholders
$E_n \geq 40$	Declared shift	Time-uniformly valid rejection at level α	Execute scale/hold/stop decision; re-anchor monitor
Instrument change ($M_t \neq M_{t-1}$)	—	Measurement intervention, not diffusion evidence	Exclude wave; re-baseline; document break

329 BTOS case.

330 3.6. Decision protocol

331 Table 1 maps monitor states to actions, separating *statistical* thresholds
332 (fixed by α and the e-process) from *managerial* ones (encoding the asymme-
333 try of scaling too early versus too late, and settable per organization).

334 3.7. Comparators

335 Three baselines represent current monitoring practice, each applied at
336 every release to the naively spliced series (legacy waves followed by revised-

wording waves, as a dashboard unaware of the redesign would splice them):
 (a) a repeated two-sample z -test comparing the latest four waves with the
 preceding four, using the design-based standard errors, uncorrected at the
 5% level; (b) the same with a Bonferroni correction over the number of looks
 made so far; and (c) a moving-average crossing rule (3-wave minus 8-wave
 means changing sign), the de facto dashboard heuristic. For each we record
 the first alarm, the number of alarm waves and of distinct alarm episodes
 over the 48 possible looks, and behavior at the November 2025 redesign.

4. Data

4.1. *The Business Trends and Outlook Survey, 2023–2025*

The BTOS is an experimental data product of the U.S. Census Bureau
 designed to measure business conditions at high frequency. The sample
 comprises approximately 1.2 million nonfarm employer businesses divided
 into six rotating panels of roughly 200,000; each panel reports once every
 twelve weeks over a year, and collection occurs every two weeks, yielding
 a continuous biweekly national series (U.S. Census Bureau, 2025, 2026b).
 Estimates are weighted to be representative at the national, state, sector,
 and employment-size levels; the average biweekly response rate over the AI-
 content period is about 16 percent (Bonney et al., 2024), and estimates failing
 publication standards are suppressed.

Two core AI questions have run since the collection period beginning 11
 September 2023 (Bonney et al., 2024). From September 2023 through Oc-
 tober 2025 the current-use question read: *“In the last two weeks, did this
 business use Artificial Intelligence (AI) in producing goods or services? (Ex-*

361 *amples of AI: machine learning, natural language processing, virtual agents,*
362 *voice recognition, etc.)”* Beginning with the collection of 17 November 2025,
363 “in producing goods or services” was replaced by “in any of its business
364 functions,” and a supplement fielded from 17 November 2025 to 8 February
365 2026 added detail on adoption across fifteen business functions ([U.S. Census](#)
366 [Bureau, 2026b](#)). A companion question asks about expected AI use over the
367 next six months under the same wordings.

368 Our analysis dataset consists of every published national and employment-
369 size-class estimate of the two AI questions, with design-based standard errors,
370 from the survey’s public data downloads. Because the Census Bureau’s file
371 of 15 December 2025 removed the legacy-wording AI rows when introducing
372 the revised series, we splice two archived vintages of the official National
373 and Employment Size Class files—downloaded 8 December 2025 (which re-
374 tains the full legacy series) and 15 December 2025 (which carries the revised
375 series)—following the vintage documentation of [Goldschlag \(2025\)](#). The re-
376 sult is 56 national waves: 54 legacy-wording waves covering collection periods
377 ending 24 September 2023 through 5 October 2025 (periods 202319–202520),
378 and two revised-wording waves ending 30 November and 14 December 2025
379 (202524–202525). No AI estimates were published for the three transition
380 periods 202521–202523. National design-based standard errors have median
381 0.15 percentage points (range 0.08–0.23); size-class standard errors are larger,
382 with medians from 0.23 (1–4 employees) to 1.21 (250+ employees) percent-
383 age points. The exact analysis dataset (two CSV files) and replication code
384 accompany the manuscript.

385 The headline contours of the series are as follows. Under the legacy

wording, national current use rose from 3.7 percent (24 September 2023) to 4.9 percent by end-2023, slipped to 4.7 percent by 30 June 2024, resumed growth to 6.3 percent by 29 December 2024, accelerated to 8.3 percent by 20 April 2025, flattened near 8.8 percent through 10 August 2025, and closed the legacy period at 10.0 percent on 5 October 2025. Planned use ran above current use throughout, from 6.3 to 14.0 percent. Under the revised wording, current use printed 17.3 and 17.2 percent and planned use 21.1 and 20.9 percent in the two available waves—consistent with subsequently published Census reporting that the broadened measure hovered between 17 and 20 percent into 2026 ([U.S. Census Bureau, 2026a](#)).

4.2. Long-run diffusion series, 1990–2026

Two public series extend the surveillance horizon to three and a half decades. *Internet users as a share of the U.S. population*, annual, 1990–2024, from the International Telecommunication Union’s World Telecommunication/ICT Indicators database as disseminated by the World Bank (indicator IT.NET.USER.ZS) and retrieved from FRED ([World Bank, 2026](#)): the series rises from 0.79 percent in 1990 to 94.7 percent in 2024, tracing the canonical S-curve of a saturating general-purpose technology. It splices household-survey estimates with ITU and World Bank estimation methods, so—unlike the BTOS—it carries neither design-based standard errors nor strata, and it contains a conspicuous single-year discontinuity of +10.9 points between 2015 and 2016 following a decade in which measured penetration fluctuated between 69 and 75 percent. *E-commerce sales as a share of total U.S. retail sales*, quarterly and seasonally adjusted, 1999Q4–2026Q1, from the Census Bureau’s Quarterly Retail E-Commerce Sales program retrieved from FRED

411 (U.S. Census Bureau, 2026c): 106 quarters from 0.6 to 16.9 percent, including
412 the celebrated COVID-19 jump (11.9 to 16.3 percent between 2020Q1 and
413 2020Q2), the 2021–2022 partial reversal to 14.2 percent, and renewed growth
414 thereafter. Both extracts are included verbatim in the supplementary data.

415 5. Results

416 5.1. Episode 1: the takeoff and its declaration (RQ1, RQ3)

417 Figures 2 and 3 display the series and the monitors’ evidence trajectories.
418 Epoch 1 opens at the second published wave (8 October 2023) against the pre-
419 registered flat baseline. Evidence accumulates quickly through the late-2023
420 growth spurt: the upward monitor passes the amber threshold ($E \geq 2$) on
421 19 November 2023, passes 10 on 17 December 2023 ($E = 15.4$), and reaches
422 a local peak of 17.1 on 28 January 2024. It then does something a p -value
423 stream cannot do: it retreats. As measured adoption stalls through the first
424 half of 2024 (4.9 percent at end-2023 to 4.7 percent on 30 June 2024), the
425 negative increments consume accumulated wealth, and the monitor bottoms
426 at $E = 0.4$ on 7 April 2024—the evidence for a takeoff, evaluated against
427 the full history, had genuinely weakened. The monitor rebuilds through
428 the second-half-2024 recovery (6.3 percent by 29 December 2024) and the
429 early-2025 acceleration (6.0 percent on 12 January to 8.3 percent on 20 April
430 2025), and crosses the declaration threshold on **20 April 2025** with terminal
431 e-value **42.2**, after 41 monitored increments. By Ville’s inequality (4), the
432 probability that a flat latent path would ever have produced such a crossing
433 is at most 2.5 percent, no matter how many times the series was inspected
434 along the way.

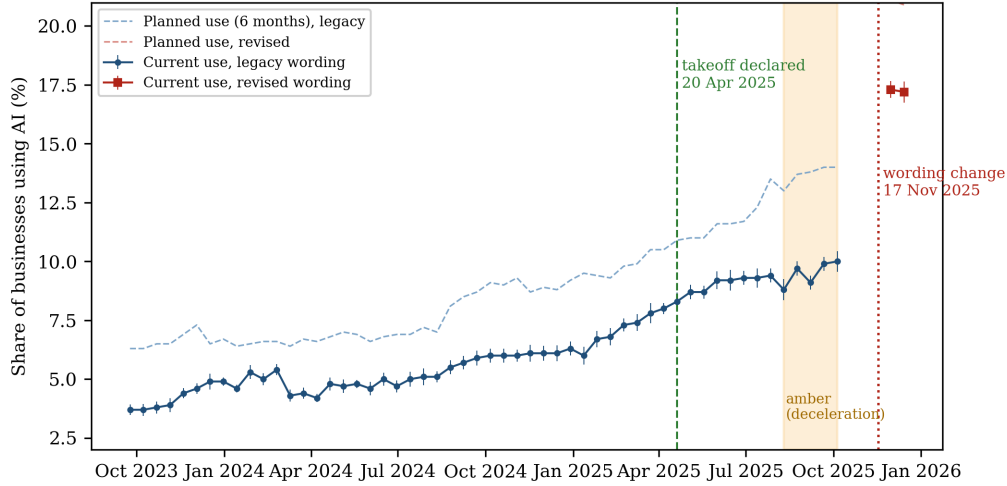


Figure 2: The BTOS national AI adoption series, September 2023–December 2025, with monitoring outcomes. Solid lines with 95% intervals show current use under the legacy (blue circles) and revised (red squares) wordings; dashed lines show planned use over the next six months. The green dashed line marks the takeoff declaration (20 April 2025); the shaded band marks the span during which the deceleration monitor held amber-or-higher evidence (10 August–5 October 2025); the red dotted line marks the question-wording intervention (17 November 2025). No AI estimates were published for the three transition periods between 5 October and 30 November 2025. Source: authors’ calculations from U.S. Census Bureau BTOS public data.

435 The declaration date is informative rather than embarrassing. The mon-
436 itor did not certify a takeoff during the 2023 spurt because, at that point, a
437 flat path punctuated by a transient could not yet be excluded at the stated
438 error rate—and the first half of 2024 showed exactly such a stall. The decla-
439 ration arrives precisely when the acceleration becomes sustained. The answer
440 to RQ1 is therefore affirmative in a specific sense: real-time detection with
441 time-uniform guarantees is feasible on official high-frequency adoption data,

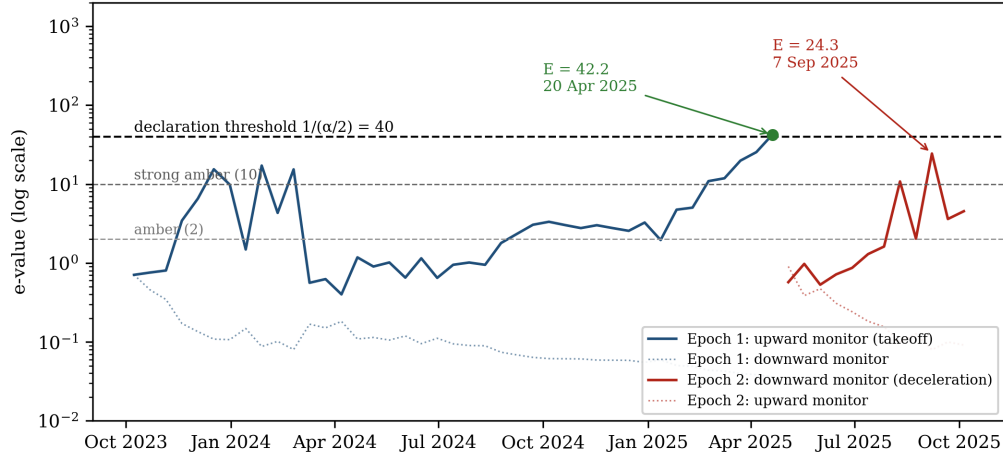


Figure 3: E-value trajectories of the national monitors (log scale). Epoch 1 tests the pre-registered flat baseline ($\beta_1 = 0$); Epoch 2 tests the anchored early-2025 growth rate ($\hat{\beta}_2 = 0.299 \pm 0.027$ pp per wave, estimated on the eight waves ending at the declaration). The upward monitor crosses the declaration threshold of 40 on 20 April 2025 ($E = 42.2$); the Epoch-2 downward monitor peaks at $E = 24.3$ on 7 September 2025 without crossing.

442 and the guarantee's price is measured restraint during ambiguous phases, not
 443 blindness.

444 5.2. Episode 2: the mid-2025 deceleration as graded evidence (RQ3)

445 Upon declaration, the monitor re-anchors: weighted least squares on the
 446 eight waves ending at the declaration (12 January–20 April 2025) fixes the
 447 Epoch-2 baseline at $\hat{\beta}_2 = 0.299 \pm 0.027$ percentage points per wave—the
 448 null that the early-2025 growth tempo continues. Epoch 2 opens on 4 May
 449 2025. Against this demanding baseline, the flattening of the series through
 450 the summer (8.3 percent in April to 8.8 percent on 10 August 2025) drives
 451 the *downward* monitor upward: it reaches amber on 10 August 2025, and
 452 peaks at $E = 24.3$ on 7 September 2025—24-to-1 evidence that growth had

453 fallen below the anchored tempo, within the strong-amber band of Table 1
454 but short of declaration. The evidence then recedes to $E = 4.5$ by 5 October
455 2025 as growth resumes (9.9 and 10.0 percent in the final legacy waves). The
456 upward monitor never exceeds 0.9 in Epoch 2.

457 Two readings of this episode matter for practice. First, the monitor
458 *quantified* the widely discussed 2025 deceleration in real time—at its peak, a
459 manager consulting the dashboard would have seen 24-to-1 odds that AI dif-
460 fusion momentum had broken—while correctly declining to certify a regime
461 change that the subsequent data contradict: the series re-accelerated within
462 weeks, and the revised-wording waves and later Census reporting show con-
463 tinued growth (U.S. Census Bureau, 2026a). Had the organization operated
464 at $\alpha = 0.10$ instead of 0.05 (declaration threshold 20), a deceleration *would*
465 have been declared on 7 September 2025 and then contradicted—a concrete
466 illustration of what the choice of α buys. Second, the moving-average heuris-
467 tic run over the same episode crossed downward on 7 September and back
468 upward on 21 September 2025 (Figure 5): a whipsaw delivering two contra-
469 dictory signals fourteen days apart, with no statement of error probability
470 attached to either.

471 5.3. Episode 3: the November 2025 redesign as a measurement event (RQ2)

472 Between the last legacy wave (10.0 percent, 5 October 2025) and the
473 first revised-wording wave (17.3 percent, 30 November 2025) the published
474 series jumps 7.3 percentage points—25.7 combined standard errors, against
475 a within-regime national history in which the median absolute standard-
476 ized movement is 0.94 and the maximum is 6.2. Because the redesign was
477 announced by the data provider, the causal layer excludes the transition in-

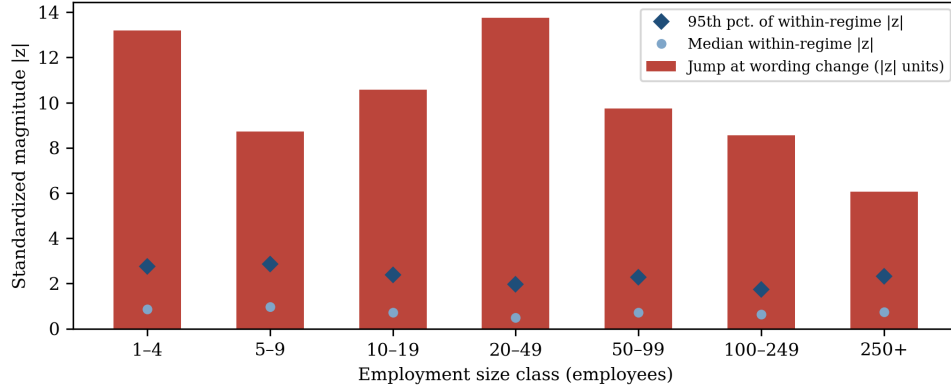


Figure 4: Cross-strata attribution diagnostic at the 17 November 2025 wording change. Bars show each employment-size class’s jump between its last legacy-wording and first revised-wording estimate, in units of the combined design-based standard error; markers show the median and 95th percentile of that class’s within-regime standardized wave-to-wave movements. Every class jumps simultaneously, by 2–7 times its within-regime 95th percentile, with a common sign—the signature of an instrument intervention, not of diffusion.

478 crement by construction, and the monitor books no diffusion evidence from
479 it. The cross-strata diagnostic of Section 3.4, applied as if the break had
480 *not* been announced, confirms the attribution decisively (Figure 4 and Ta-
481 ble 3): all seven employment-size classes jump in the same single wave, by 6.1
482 to 13.8 standard errors—between roughly 2.6 and 7.0 times their respective
483 within-regime 95th percentiles (1.74–2.85)—and all in the same direction, as
484 the mechanically broader wording implies. Nothing in the genuine takeoff
485 resembles this signature: the takeoff built over more than a year, wave by
486 wave, with movements inside or near the within-regime tail.

487 The contrast with conventional practice is stark. The naive rolling z -test,
488 applied to the spliced series as a redesign-unaware dashboard would splice it,

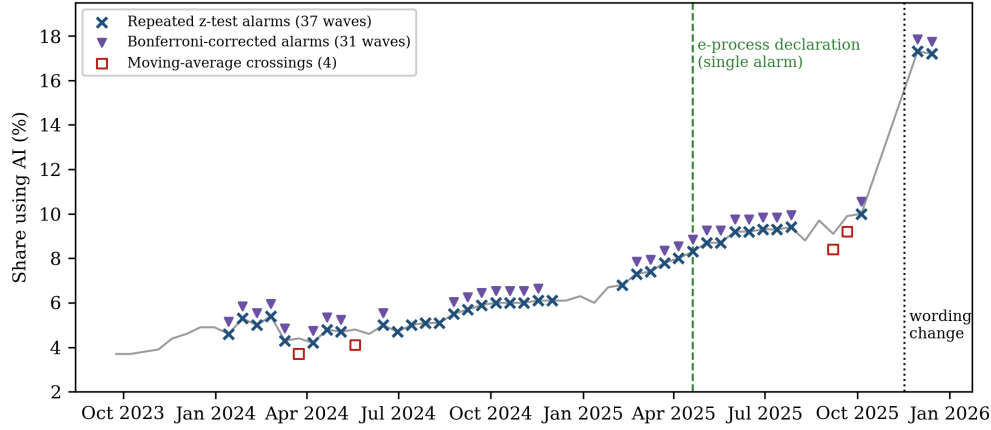


Figure 5: Alarm behavior of conventional monitoring rules over the naively spliced series, September 2023–December 2025. Crosses: repeated four-versus-four-wave z -test, uncorrected (37 alarm waves of 48 looks). Triangles: the same with Bonferroni correction over looks (31 alarm waves). Open squares: 3-versus-8-wave moving-average crossings (four crossings, including the September 2025 whipsaw). The e-process framework produces a single declaration (green dashed line) and excludes the wording change (dotted line) by design.

489 produces its largest statistic of the entire sample at the break ($|z| = 32.9$) and
 490 alarms there under both the uncorrected and Bonferroni-corrected versions—
 491 certifying, at any conventional significance level, the largest “diffusion shock”
 492 on record where none occurred. The answer to RQ2 is that the separation
 493 is achievable and cheap: it requires only that the monitoring architecture
 494 carry the instrument regime as an explicit variable and treat its changes as
 495 interventions, with the cross-strata signature available as verification when
 496 provenance is uncertain.

Table 2: Detection and alarm behavior across monitoring methods on the BTOS national AI series (48 possible looks, September 2023–December 2025).

Method	First alarm	Alarm waves	Distinct episodes	Alarms at redesign
Causal e-process ($\alpha = 0.05$)	20 Apr 2025 ^a	1	1	0 (excluded by design)
Repeated z -test, uncorrected	14 Jan 2024	37	5	yes ($ z = 32.9$)
Repeated z -test, Bonferroni	14 Jan 2024	31	6	yes
Moving-average crossing (3, 8)	24 Mar 2024	4	3	no crossing

^aDeclaration with terminal e-value 42.2; graded evidence available continuously beforehand (amber from 19 November 2023, strong amber from 17 December 2023, trough 0.4 on 7 April 2024). Moving-average crossing dates: 24 March 2024, 19 May 2024, 7 September 2025, 21 September 2025.

5.4. Comparison with conventional monitoring (RQ3)

Table 2 and Figure 5 summarize the comparison, and the honest reading cuts both ways. The repeated z -tests “detect” growth on 14 January 2024, fifteen months before the e-process declaration—but the same rule then alarms on 37 of 48 looks (31 after Bonferroni), including through the first-half-2024 stall and at the November 2025 redesign. A rule that is almost always alarming cannot date a regime change, cannot distinguish diffusion from measurement, and offers no interpretable error statement under open-ended monitoring: the uncorrected version’s family-wise guarantee is void, and the Bonferroni version’s guarantee, while formally valid look-by-look, is purchased by a level that shrinks toward zero as monitoring continues, with its 31 alarms demonstrating that in this high-precision setting the correction restrains nothing that matters. The moving-average rule is quiet by comparison but delivers whipsaws (two crossings in 2024 during the stall-

Table 3: Employment-size-class series and monitoring outcomes. “Max E^+ ” is the maximum of the takeoff (Epoch-1) upward monitor over the legacy period; no class reaches the declaration threshold of 40, reflecting larger stratum-level standard errors (median s.e. in percentage points shown). Jump columns give the movement between the last legacy-wording and first revised-wording estimates in percentage points and in combined standard errors, against the 95th percentile of within-regime standardized movements.

Size class	Sep 2023	Oct 2025	Nov 2025 ^r	Med. s.e.	Max E^+	Jump (pp)	Jump ($ z $)	p95 wit
1–4	4.3	11.0	17.0	0.23	11.4	6.0	13.2	2.7
5–9	3.0	8.6	15.5	0.29	4.0	6.9	8.7	2.8
10–19	2.6	8.1	17.8	0.32	10.7	9.7	10.6	2.3
20–49	2.6	7.0	18.7	0.36	1.8	11.7	13.8	1.9
50–99	2.4	9.7	20.7	0.57	4.0	11.0	9.7	2.2
100–249	2.5	8.8	23.7	0.92	1.3	14.9	8.6	1.7
250+	3.1	13.0	30.5	1.21	12.9	17.5	6.1	2.3

Entries in percent of businesses except where noted. ^rFirst revised-wording estimate (30 November 2025). Source: authors’ calculations from U.S. Census Bureau BTOS public data (Employment Size Class files, vintages of 8 and 15 December 2025).

511 recovery, two in September 2025) with no error control at all. The e-process
512 trades earliness for validity and interpretability: one declaration, dated, with
513 a terminal evidence ratio; a continuous graded evidence path before it; cor-
514 rect restraint during the plateau; and structural immunity to the instrument
515 change. Which side of that trade a decision-maker should take depends on
516 the cost of false regime declarations—and Section 6 argues that for scaling
517 decisions in technology management, those costs are high.

518 *5.5. Heterogeneity across firm-size classes*

519 Table 3 reports the size-class analysis. Levels replicate the familiar gradient—
520 by the revised measure, adoption runs from 17.0 percent among the smallest
521 firms to 30.5 percent among firms with 250 or more employees, and the
522 revised-wording jump is largest in absolute terms for large firms (17.5 per-
523 centage points), whose multi-function operations the legacy wording most
524 severely under-counted. Stratum-level monitors, facing standard errors four
525 to eight times the national ones, accumulate takeoff evidence that peaks
526 in amber territory (maximum E^+ of 11.4–12.9 for the smallest and largest
527 classes, weaker in between) without any stratum-level declaration in-sample—
528 a transparent illustration that the framework’s detection power is governed
529 by the precision of the stream it watches, and that national-level surveillance,
530 which pools that precision, is where declarations are earned. For the attribu-
531 tion diagnostic, the strata are decisive in the other direction, as Section 5.3
532 showed: it is exactly because every class jumps at once, far outside its own
533 history, that the redesign is unmistakable as a measurement event.

534 *5.6. The long view: three diffusion waves, 1990–2026*

535 Figure 6 and Table 4 place the AI episode in its evolutionary context. Run
536 with the plug-in-variance monitor of Section 3.5, the surveillance recovers the
537 punctuated record of U.S. digital diffusion as a sequence of dated declara-
538 tions. For the internet, the takeoff is declared in 1995—the year commercial
539 access became mainstream—against the pre-registered flat baseline; against
540 the anchored boom-era growth rate of 6.75 points per year (1996–2003), the
541 downward monitor declares the onset of saturation in 2008; and no further
542 shift is declared through 2024 against the ensuing plateau-era baseline. For

e-commerce the sequence is denser: takeoff declared in 2004Q1; a genuine acceleration in the growth *rate*—from about 0.10 to about 0.32 points per quarter—declared in 2018Q1; the COVID-19 lockdown shock declared in the very quarter it occurred, 2020Q2, with decisive evidence; and, against the post-pandemic baseline of *declining* share (-0.11 per quarter, anchored on 2020Q3–2022Q2), the resumption of growth declared in 2023Q1. The monitor is quiet where the series is quiet: no declaration stands open at the end of either sample.

Two lessons carry back to the paper’s core argument. First, the COVID-19 episode furnishes the natural counterpart to the November 2025 BTOS redesign: both are large single-period jumps, but the 2020Q2 shock is a genuine diffusion event—externally corroborated, persistent in level, and occurring within an unchanged measurement instrument—whereas the 2025 jump is pure instrument. A monitoring architecture must be able to declare the first and absorb the second, and this one does. Second, the internet series contains the cautionary intermediate case: a $+10.9$ -point jump in a single year (2015–2016) after a decade-long statistical plateau, a profile that under the diagnostic of Section 3.4 is highly suspect as a measurement artifact—the series splices household-survey estimates with ITU and World Bank estimation methods (World Bank, 2026)—but that cannot be adjudicated, because no design-based errors, strata, or documented redesign dates accompany it. In our epoch structure the jump falls inside the 2009–2016 anchoring window and therefore contaminates the plateau-era baseline rather than triggering a declaration; an analyst without the measurement layer would have been left to guess. The comparison across the three panels of Figure 6 is thus also a

568 comparison across measurement regimes—annual splices, quarterly designed
569 surveys, biweekly designed surveys with published errors and strata—and it
570 makes concrete how much inferential reach the improvement in measurement
571 infrastructure has purchased.

572 *5.7. The intermediate waves: data mining, cloud, big data, data science, and*
573 *deep learning*

574 Between the internet wave and the AI wave lie the technologies through
575 which the analytics era actually arrived—data mining in the 1990s, cloud
576 computing from 2006, big data and data science from the early 2010s, and
577 the deep-learning breakout of 2012–2015. Figure 7 puts all of them in one
578 picture, and it does so under an honesty constraint that is itself a find-
579 ing: *none* of these waves was ever tracked by an official penetration series.
580 No statistical agency asked businesses, at any frequency, whether they used
581 data mining in 1997, cloud infrastructure in 2009, or deep learning in 2016;
582 the first designed, recurring, error-quantified adoption instrument for any of
583 these general-purpose capabilities is the 2023 BTOS AI question set analyzed
584 in this paper. The intermediate waves are therefore dated in panel (a) by
585 canonical landmarks—the founding KDD conference and framework (Fayyad
586 et al., 1996), the EC2 launch and the Berkeley cloud retrospective (Armbrust
587 et al., 2010), the big-data research agenda (Chen et al., 2012), the codification
588 of data science (Provost and Fawcett, 2013), AlexNet and the deep-learning
589 consolidation (Krizhevsky et al., 2017; LeCun et al., 2015), and the Chat-
590 GPT release—rather than by measured curves, and we deliberately decline to
591 substitute proxies such as search-interest or bibliometric indices for adoption
592 shares within the monitoring framework, since their measurement processes

593 (query-share renormalization, corpus growth, indexing-policy changes) are
594 exactly the kind of undocumented instrument drift the framework exists to
595 quarantine.

596 Read jointly, the two panels tell an evolutionary story with a measurement
597 moral. The discourse waves cluster in the long gap between the internet’s
598 declared saturation onset (2008) and the AI takeoff (2025)—the period in
599 which the capability stack (storage, elastic compute, learning algorithms) was
600 assembled that the 2023–2025 AI diffusion then drew upon; the e-commerce
601 acceleration declared in 2018Q1 is the one place where an intermediate wave
602 leaves a footprint in a measured series, consistent with the maturation of
603 cloud-era retail infrastructure in the mid-2010s. That the AI wave is the
604 first to be observable biweekly, with design-based errors and strata, is why
605 the monitoring guarantees of this paper became possible at all—and why,
606 prospectively, the framework can be in place from a technology wave’s first
607 measured wave rather than reconstructed after the fact.

608 5.8. *Robustness*

609 Two sensitivity checks probe the construction. First, the prior scale:
610 re-running the full surveillance with $\tau = 0.5$ and $\tau = 2$ leaves the takeoff
611 declaration on 20 April 2025 for $\tau \in \{0.5, 1\}$ (terminal e-values 58.8 and
612 42.2) and moves it one wave later, to 4 May 2025, for $\tau = 2$ ($E = 45.8$);
613 the qualitative Epoch-2 pattern is unchanged. Second, the sharp attribution
614 of within-regime dispersion to sampling error: inflating every σ_t by a factor
615 $\kappa = 1.5$ —equivalent to positing that half again as much wave-to-wave noise is
616 non-informational—prevents any declaration in-sample (the takeoff monitor
617 peaks at 5.8). The framework’s conclusions are thus robust to the prior but,

618 as they should be, sensitive to the variance model: the design-based stan-
619 dard errors are the published, defensible quantification of sampling noise for
620 this survey, and analysts who believe substantial idiosyncratic latent wobble
621 exists can encode that belief in κ and accept correspondingly later detection.
622 We regard the transparency of that dial as a feature: it makes explicit an
623 assumption that rolling t -tests bury.

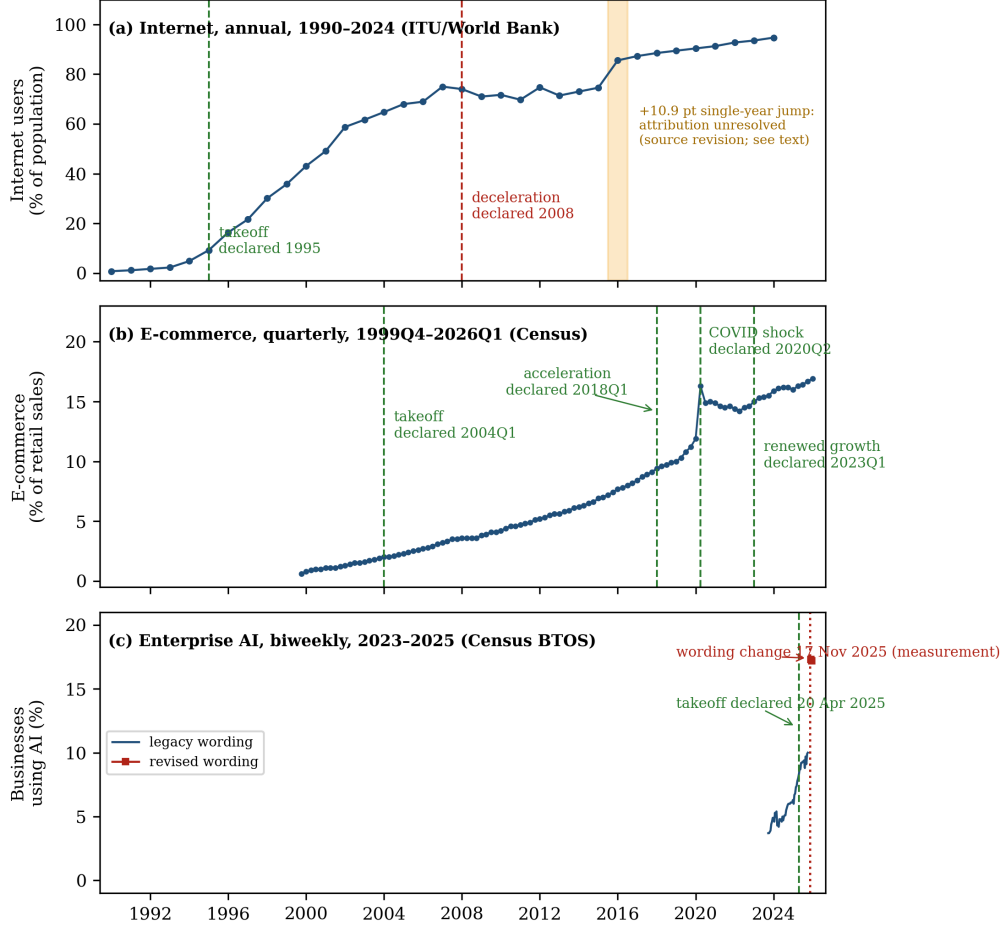
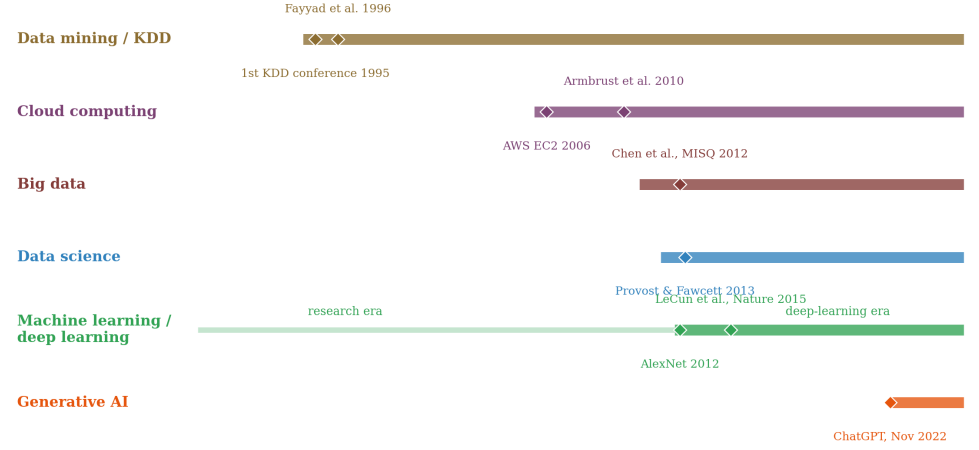


Figure 6: Three decades of U.S. digital technology diffusion under anytime-valid surveillance. (a) Internet users as a share of population, annual (ITU/World Bank via FRED); (b) e-commerce share of retail sales, quarterly (Census via FRED); (c) enterprise AI use, biweekly (Census BTOS; this paper’s core application). Green dashed lines mark declared regime shifts; the red dashed line in (a) marks the declared deceleration; the red dotted line in (c) marks the excluded measurement intervention; the shaded band in (a) marks the 2016 single-year discontinuity whose attribution the available metadata cannot settle. Panels (a) and (b) use the plug-in-variance monitor of Section 3.5; panel (c) uses the exact design-based monitor.

Table 4: Declared regime shifts across three diffusion waves, 1990–2026. The internet and e-commerce rows use the plug-in-variance monitor (approximate validity; Section 3.5); the enterprise-AI rows use the exact design-based monitor. Baseline slopes are in percentage points per period of the respective series (year, quarter, biweekly wave). E-values above 10^4 are reported as decisive.

Series and epoch baseline slope	Declared shift	Direction	Terminal E
<i>Internet (annual)</i>			
0 (pre-registered)	1995	up (takeoff)	$> 10^4$
6.75 (anchored 1996–2003)	2008	down (saturation onset)	192.7
1.46 (anchored 2009–2016)	none by 2024	—	0.7
<i>E-commerce (quarterly)</i>			
0 (pre-registered)	2004Q1	up (takeoff)	59.0
0.10 (anchored 2004–2006)	2018Q1	up (acceleration)	52.4
0.32 (anchored 2018–2020)	2020Q2	up (COVID-19 shock)	$> 10^4$
−0.11 (anchored 2020–2022)	2023Q1	up (renewed growth)	79.1
0.13 (anchored 2023–2025)	none by 2026Q1	—	1.3
<i>Enterprise AI (biweekly)</i>			
0 (pre-registered)	20 Apr 2025	up (takeoff)	42.2
0.299 (anchored Jan–Apr 2025)	none by Dec 2025	(deceleration peaked at 24.3)	4.5

(a) Technology waves named in discourse, landmark-dated (no official penetration series exist)



(b) Regime shifts declared by the anytime-valid monitor (Table 4)

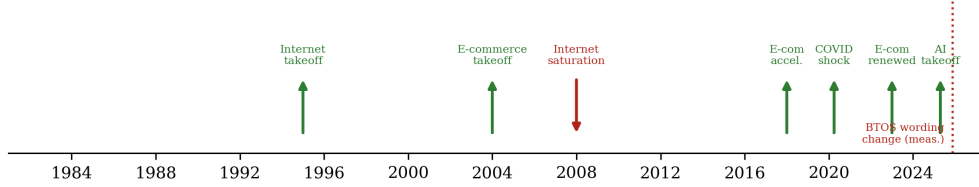


Figure 7: The technology waves of the analytics era, 1990–2026, aligned with the measured record. Panel (a) places the successive waves of technology discourse—data mining and knowledge discovery in databases (Fayyad et al., 1996), cloud computing (Armbrust et al., 2010), big data (Chen et al., 2012), data science (Provost and Fawcett, 2013), the deep-learning breakout within machine learning (Krizhevsky et al., 2017; LeCun et al., 2015), and generative AI—on a common timeline, each dated by canonical, citable landmarks (diamonds) rather than by penetration series, because no official adoption-share statistics were ever produced for these intermediate waves. Panel (b) repeats, on the same axis, the regime shifts declared by the anytime-valid monitor on the three measured series of Table 4. Landmark dates: first ACM KDD conference (1995) and the KDD process framework (Fayyad et al., 1996); Amazon EC2 launch (2006) and the Berkeley view of cloud computing (Armbrust et al., 2010); the MIS Quarterly big-data agenda (Chen et al., 2012); the codification of data science (Provost and Fawcett, 2013); AlexNet (Krizhevsky et al., 2017) and the Nature review of deep learning (LeCun et al., 2015); and the public release of ChatGPT (November 2022), the proximate trigger of the generative-AI wave (Bick et al., 2025).

624 6. Discussion

625 6.1. Implications for technology and innovation management research

626 Methodologically, the results argue that the arrival of streaming adoption
627 data requires a matching change in inferential practice. Any TIM study that
628 monitors an ongoing series—of adoption, patenting, platform activity—and
629 reports significance at a data-dependent stopping point inherits the peeking
630 problem, and Table 2 shows how badly it bites on real official data: even Bon-
631 ferroni correction leaves a rule that alarms on nearly two-thirds of looks and
632 certifies a survey redesign as the largest diffusion event on record. Anytime-
633 valid tools remove the problem at negligible implementation cost—equation
634 (3) is one line—and change what a “finding” is: instead of a curve inflec-
635 tion estimated in hindsight, the analyst obtains a dated, auditable evidence
636 trajectory recording exactly how much support a regime shift had at every
637 moment. For the evolutionary perspective animating this special issue, that
638 trajectory is the natural object: takeoffs, stalls, and punctuations become
639 detected events with attached error guarantees, and the 2023–2025 history
640 of enterprise AI—spurt, stall, sustained acceleration, plateau, resumption—is
641 recovered here as precisely such a sequence.

642 Theoretically, treating measurement redesigns as causal interventions re-
643 frames the survey-comparability debate. The dispersion of AI adoption esti-
644 mates across instruments (Crane et al., 2025) and the near-doubling of the
645 BTOS measure under broader wording (Bick, 2026) are usually offered as
646 reasons to distrust any single series. In our framework they are identified
647 components of the observation function g in (1), and the November 2025
648 episode becomes a validation asset: a monitor that cannot absorb a known

instrument intervention should not be trusted to interpret unknown ones. The cross-strata signature developed here—synchronization, extremity relative to each stratum’s own history, mechanical sign coherence—gives applied researchers a portable screening rule for the redesigns that high-frequency programs will keep producing.

6.2. *Implications for practice*

For managers, the immediate application is internal. Enterprise AI deployments generate adoption telemetry—weekly active users, function-level usage, ticket volumes—that is reviewed continuously and acted upon under exactly the conditions that invalidate conventional tests. The protocol of Table 1 transfers directly: define the baseline path for a pilot, run upward and downward monitors at a chosen α , treat tool migrations and instrumentation changes as measurement interventions, and tie scale/hold/stop decisions to evidence thresholds. The Epoch-2 episode is the template for the hardest managerial call: when a promising deployment appears to stall, the monitor distinguishes “24-to-1 evidence of a momentum break—prepare contingencies” from “regime change certified—act,” and in this instance the distinction saved the (hypothetical) decision-maker from killing a trajectory that resumed within weeks. For analysts and policymakers consuming official statistics, the framework supplies a disciplined answer to the recurring question of whether the latest print signals genuine change, with error properties invariant to how often the question is asked.

671 *6.3. Limitations and future research*

672 Five limitations bound the claims. First, the public BTOS tabulations are
673 aggregates from a rotating panel: the monitor tracks population-level rates,
674 and firm-level regime dynamics—entry into and exit from AI use—are observ-
675 able only through restricted microdata access, a natural Federal Statistical
676 Research Data Center extension. Second, the adoption measure is a binary
677 self-report; supplements measuring use across business functions ([U.S. Cen-
678 sus Bureau, 2026b](#)) invite multivariate e-processes over adoption depth, not
679 just incidence. Third, the sharp null attributes all within-regime dispersion to
680 design-based sampling error; the κ -robustness analysis shows detection tim-
681 ing depends materially on that attribution, and developing e-processes with
682 data-driven yet valid overdispersion allowance is an open problem. Fourth,
683 the attribution diagnostic for unannounced breaks is a screening rule with
684 a causal rationale rather than an identified estimator; formal identification
685 under richer instrument models would connect this work more deeply to the
686 invariance literature. Fifth, the exact guarantees cover one technology in one
687 country through December 2025, with only two post-redesign waves—too few
688 to re-anchor the revised-regime monitor within sample; the 1990–2026 ret-
689 rospective extends the surveillance logic across technologies but only with
690 approximate, plug-in-variance validity and at coarse temporal resolution, so
691 replications on the extending BTOS series, on the Eurostat enterprise ICT
692 surveys, and on other economies’ high-frequency programs remain the path
693 to a comparative evolutionary evidence base with exact error control.

694 7. Conclusion

695 Technology adoption is now measured as a stream, but inference about
696 it is still largely conducted as if data collection ended yesterday. This pa-
697 per contributes a monitoring framework in which evidence about diffusion
698 regime shifts is anytime-valid—robust to continuous inspection and optional
699 stopping—and causally insulated from changes in the measurement appa-
700 ratus itself. On the biweekly U.S. Census series for enterprise AI use, the
701 framework declared the diffusion takeoff on 20 April 2025 at 42-to-1 evidence,
702 held the mid-2025 deceleration at a peak of 24-to-1 without a false declara-
703 tion, and absorbed a 26-standard-error survey redesign that conventional
704 tests certify as the largest diffusion shock on record. Extended retrospec-
705 tively across three decades, the same logic recovers the evolutionary record
706 of U.S. digital diffusion—the internet’s 1995 takeoff and 2008 saturation on-
707 set, e-commerce’s 2004 takeoff, 2018 acceleration, 2020 pandemic shock, and
708 2023 re-acceleration—as a sequence of dated, directionally signed declara-
709 tions, with the 2016 internet-series discontinuity standing as a reminder of
710 what is lost when measurement travels without metadata. For technological
711 forecasting as a planning discipline, the shift on offer is from a curve esti-
712 mated once, after the fact, to a monitor that is live for the entire life of the
713 technology it watches—turning the forecast itself into an instrument that
714 can be acted on the day the evidence, not the calendar, says it is time to.
715 The broader proposal to the field is a change of stance: from fitting diffu-
716 sion curves after the fact to running always-on, error-controlled surveillance
717 of how social technologies evolve through the economy—and acting on that
718 evidence with guarantees that survive the way evidence is actually consumed.

719 Declaration of competing interest

720 The authors declare that they have no known competing financial inter-
721 ests or personal relationships that could have appeared to influence the work
722 reported in this paper.

723 Data availability

724 All data are public U.S. Census Bureau Business Trends and Outlook Sur-
725 vey statistics (https://www.census.gov/hfp/btos/data_downloads). The
726 long-run series are retrieved from FRED: Internet users for the United States
727 (ITNETUSERP2USA; <https://fred.stlouisfed.org/series/ITNETUSERP2USA>)
728 and E-Commerce Retail Sales as a Percent of Total Sales (ECOMPCTSA;
729 <https://fred.stlouisfed.org/series/ECOMPCTSA>). The exact analysis
730 datasets (four CSV files: the spliced BTOS national and employment-size-
731 class series with design-based standard errors and vintage provenance, and
732 the two long-run series as retrieved) and replication scripts reproducing ev-
733 ery number, table, and figure in the paper are provided as supplementary
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738 Appendix A. The one-sided mixture e-process

739 *Derivation of (3).* Under $H_0^{(e)}$ in (2), the standardized increments $z_1, z_2, \dots$
740 are i.i.d. $\mathcal{N}(0, 1)$. For a fixed drift δ , the likelihood ratio of $\mathcal{N}(\delta, 1)$ against

741 $\mathcal{N}(0, 1)$ after n observations is $L_n(\delta) = \exp(\delta S_n - n\delta^2/2)$ with $S_n = \sum_{s \leq n} z_s$,
742 a nonnegative martingale with $L_0 = 1$. Mixing over the half-normal prior
743 $\pi_\tau(\delta) = \frac{2}{\tau} \varphi(\delta/\tau) \mathbf{1}\{\delta > 0\}$ preserves the martingale property (Fubini), giving
744 $E_n^+ = \int_0^\infty L_n(\delta) \pi_\tau(\delta) d\delta$. Completing the square with $a_n = n + \tau^{-2}$,

$$E_n^+ = \sqrt{\frac{2}{\pi\tau^2}} \int_0^\infty \exp\left(-\frac{a_n}{2}\left(\delta - \frac{S_n}{a_n}\right)^2 + \frac{S_n^2}{2a_n}\right) d\delta = \frac{2}{\tau\sqrt{a_n}} \Phi\left(\frac{S_n}{\sqrt{a_n}}\right) \exp\left(\frac{S_n^2}{2a_n}\right),$$

745 which is (3); at $n = 0$, $a_0 = \tau^{-2}$ and $E_0^+ = 2\Phi(0) = 1$.

746 *Validity under the composite one-sided null..* Suppose more generally that
747 $z_t \mid \mathcal{F}_{t-1} \sim \mathcal{N}(\mu_t, 1)$ with $\mu_t \leq 0$ (drift no greater than the baseline). For
748 any $\delta > 0$, $\mathbb{E}[e^{\delta z_t - \delta^2/2} \mid \mathcal{F}_{t-1}] = e^{\delta\mu_t} \leq 1$, so each $L_n(\delta)$ is a nonnegative
749 supermartingale, and hence so is the mixture E_n^+ . Ville's inequality for non-
750 negative supermartingales then gives $\mathbb{P}(\exists n : E_n^+ \geq 1/\alpha') \leq \alpha'$, establishing
751 (4) with $\alpha' = \alpha/2$; the same argument applies to E_n^- under $\mu_t \geq 0$, and
752 running both monitors controls the per-epoch false-declaration probability
753 at α by a union bound.

754 Appendix B. Implementation parameters

755 All computations use only information available at each wave, preserving
756 predictability. Defaults: prior scale $\tau = 1$; level $\alpha = 0.05$ split equally across
757 the two one-sided monitors (declaration threshold 40); anchoring window
758 $w = 8$ waves with weighted-least-squares slope estimation (weights $1/\hat{\sigma}_t^2$) and
759 the slope's standard error added to the increment variance as in (2); Epoch-
760 1 baseline pre-registered at $\beta_1 = 0$; pre-announced instrument-transition
761 increments excluded from all monitors. Comparators: two-sample z -tests

on four-versus-four-wave windows using design-based standard errors, uncorrected at the two-sided 5% level or Bonferroni-corrected by the running number of looks; moving-average rule with 3- and 8-wave windows. Retrospective (plug-in) runs: MAD window $W = 8$; floors $\phi = 0.8$ (internet, per year) and $\phi = 0.08$ (e-commerce, per quarter); residuals seeded from each anchoring window; e-values capped at 10^4 for reporting. The replication scripts (`reproduce_results.py` for the BTOS application; `reproduce_retro.py` for the 1990–2026 retrospective) reproduce all reported values from the supplementary CSV files.

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